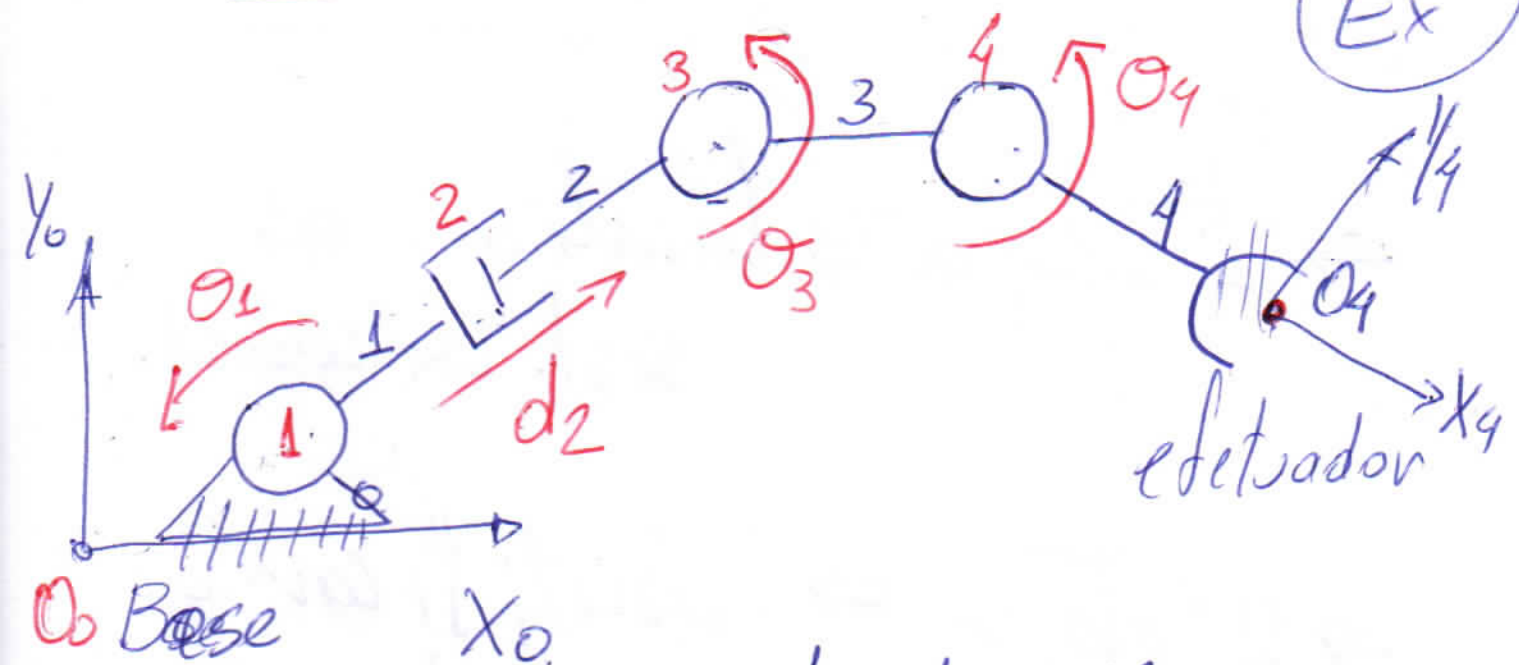


9/12

①

Cinemática direta de robôs

Ex



$z \rightarrow e \rightarrow 0$

$$j = 1 \dots n$$

$n=4$

$$ebs = 0 \dots n$$

$$\vec{q} = \begin{bmatrix} \theta_1 \\ d_2 \\ \theta_3 \\ \theta_4 \end{bmatrix}$$

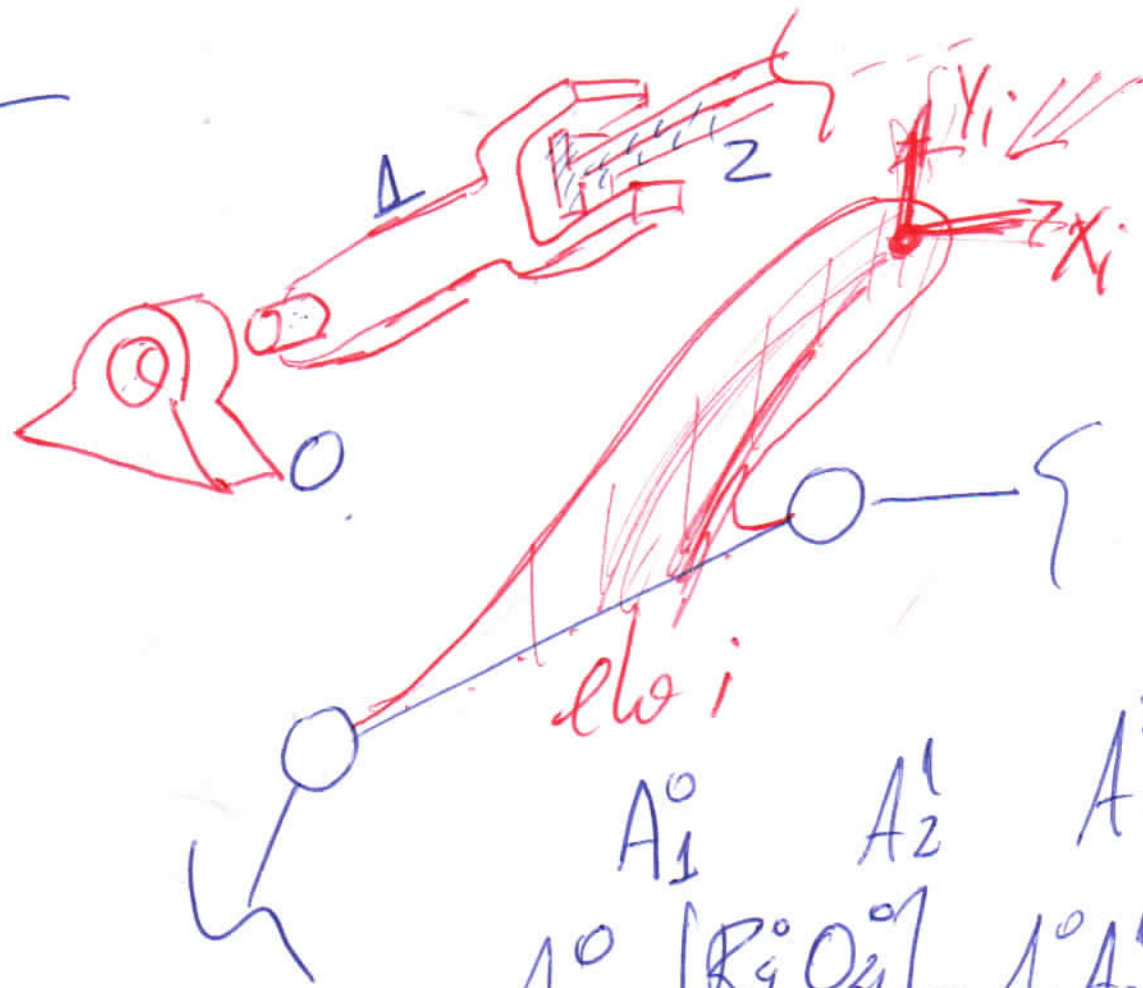
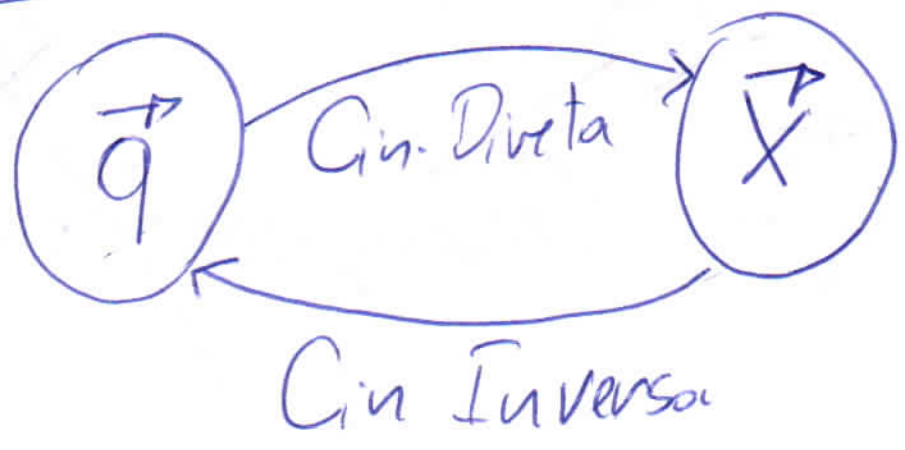
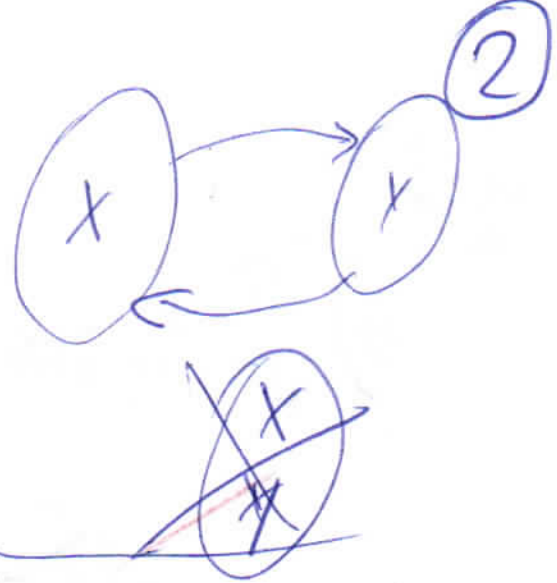
$$\vec{x} = \begin{bmatrix} x_e \\ y_e \\ \phi_z \end{bmatrix}$$

espaço vetorial
das juntas

espaço vetorial
do efetuator no
cartesiano

9/12

$$\begin{cases} Y_1 = a_1 X_1 + b_1 X_2 \\ Y_2 = a_2 X_1 + b_2 X_2 \end{cases}$$



$$A_4^0 = \begin{bmatrix} R_4^0 & O_{41}^0 \\ 0 & 1 \end{bmatrix} = \overrightarrow{A_1^0 A_2^1 A_3^2 A_4^3}$$

9/12

③

Convenção de Denavit-Hartenberg

$\Rightarrow$ 1ª etapa $\Rightarrow$ localizar os
sist. de coord.

$\Rightarrow$ 2ª etapa $\Rightarrow$ Identificar os
parâmetros DH

$$\hookrightarrow A_i^{i-1}$$

1ª etapa = localizar os sist. coord.

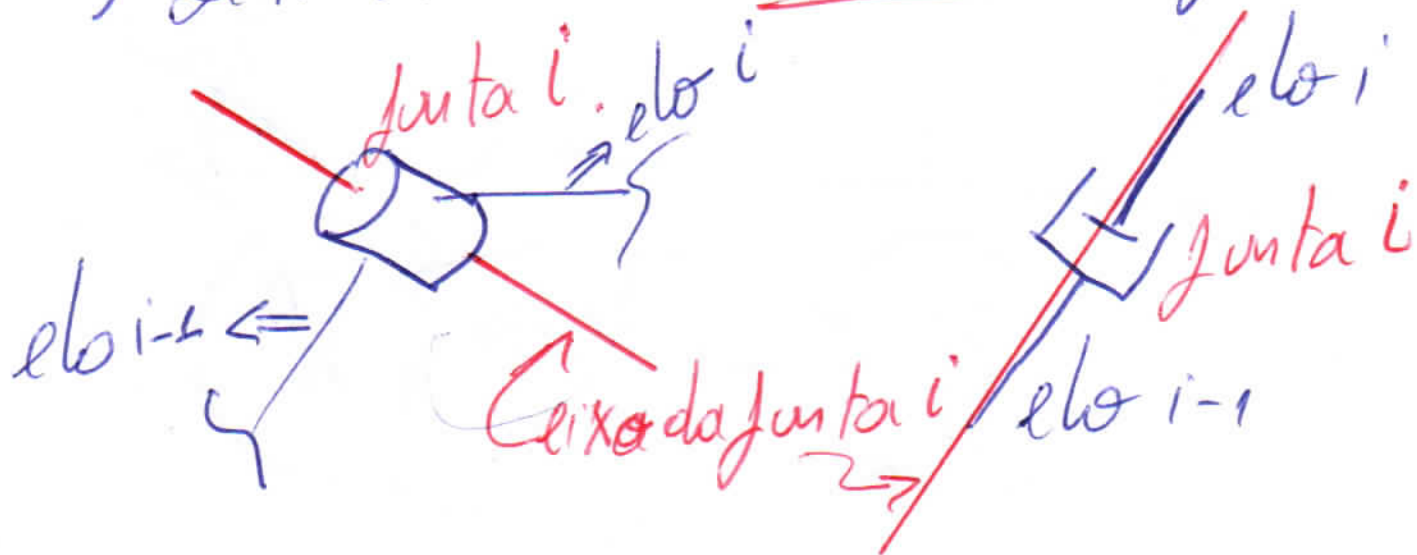
a) enumerar elos e juntas

$$\begin{array}{l} \hookrightarrow 1 \dots n \\ \hookrightarrow 0 \dots n \\ \hookrightarrow \text{base} \end{array}$$

9/12

④

b) Determinar os eixos das juntas



c) Associar Z_{i-1} ao eixo da junta i

d) Determinar a normal comum entre os eixos das juntas i e $i-1$

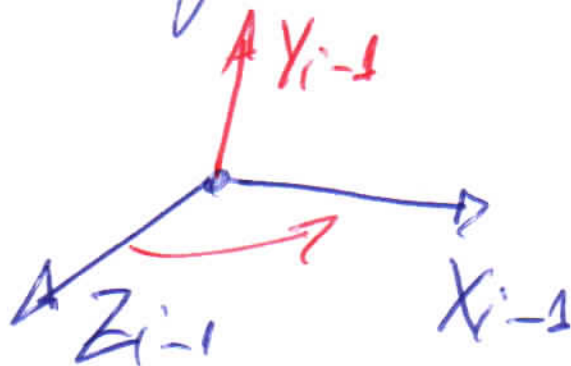
g/12

⑤

$\Rightarrow O_{i-1} \Rightarrow$ Cruzamento do eixo da junta i com a normal comum entre a junta i e $i-1$

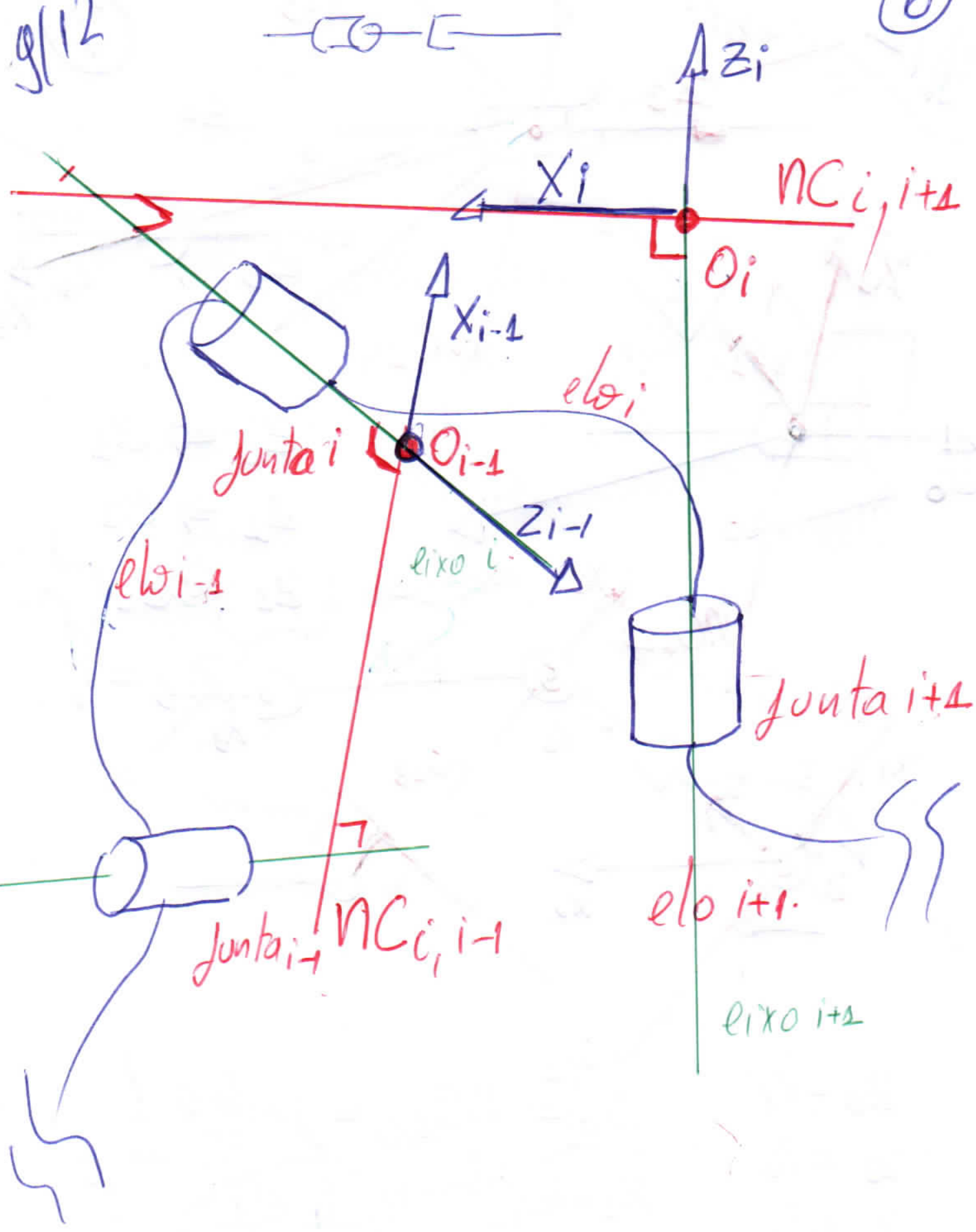
$\Rightarrow X_{i-1} \Rightarrow$ Sobre a normal comum entre a junta i e $i-1$

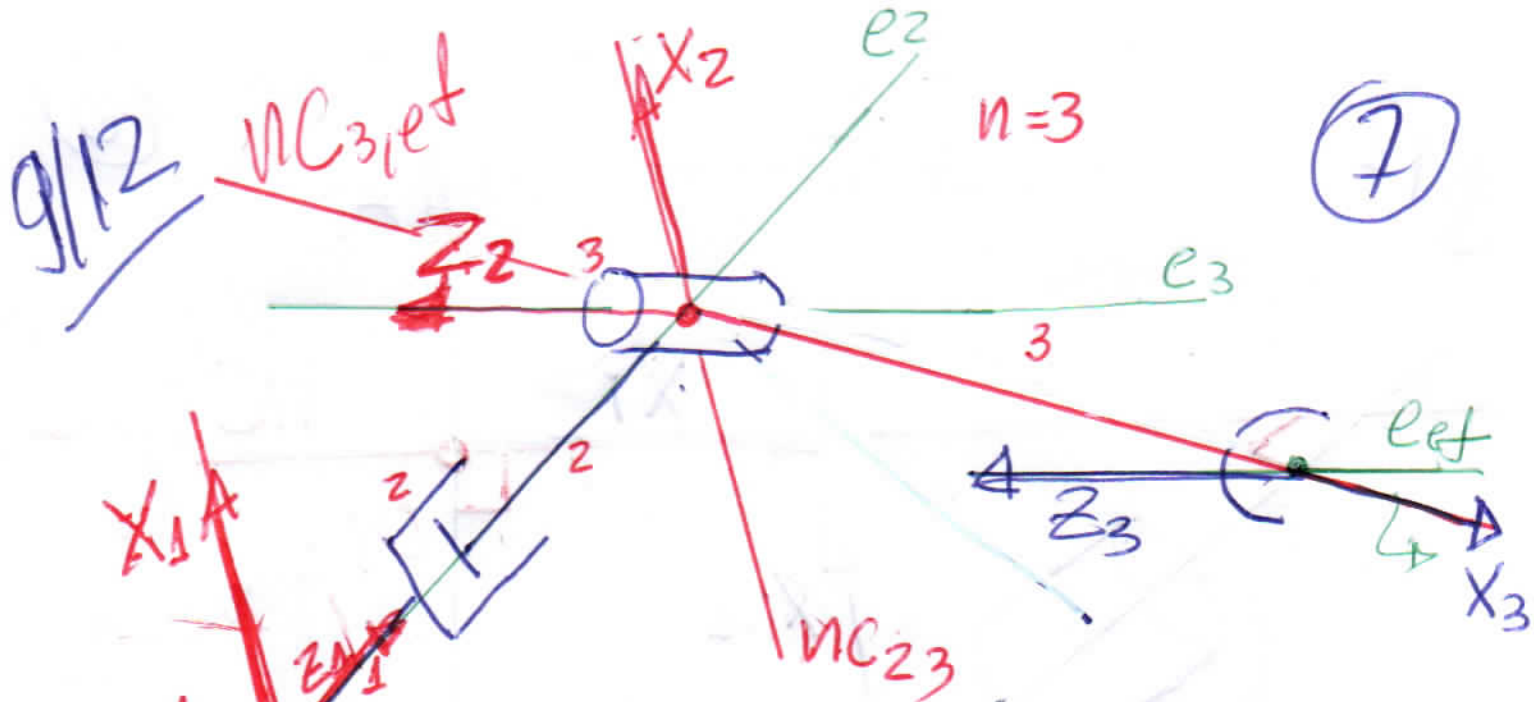
* $Y_{i-1} \Rightarrow$ regra da mão direita



9/12

⑥



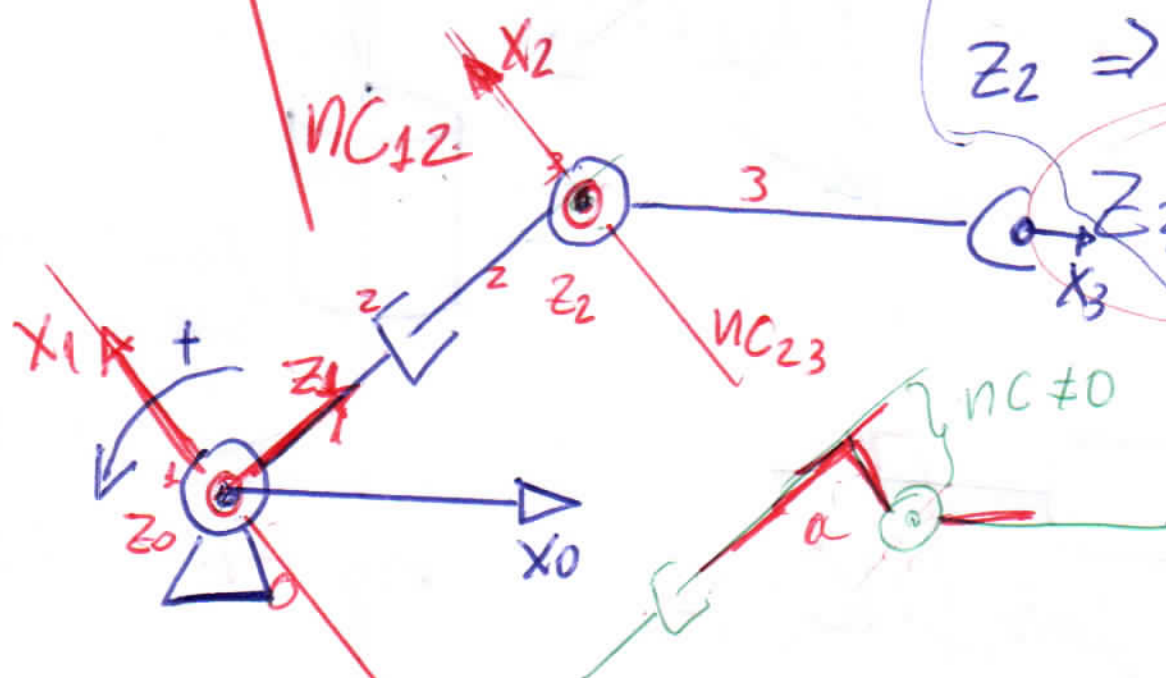


$$Z_0 \Rightarrow e_1$$

$$Z_1 \Rightarrow e_2$$

$$Z_2 \Rightarrow e_3$$

$$Z_3 = ?$$



$$Z_0 = e_1$$

$$Z_1 = e_2$$

$$Z_2 = e_3$$

$$Z_3 = ?$$

$$NC_{12}$$

$$X_0 = NC_{1,0} = \text{junto 0?}$$

$$X_1 = NC_{12}$$

$$X_2 = NC_{23}$$

$$X_3 = NC_{3,4} \Rightarrow \text{junta 4} \Rightarrow ? \Rightarrow \#$$

$$e_1$$

$$NC \neq 0$$

8/12

8

$$Z_3 \Rightarrow Z_n$$

└─> eixo do efetuator

* $\Rightarrow$ // ao eixo da junta n

* = alinhado ao do n

$$Z_3 = C_{ef} \Rightarrow X_3 \Rightarrow nC_{3,ef}$$

$$X_n = nC_{n,ef}$$

